

PDC Assignment 3 — Distributed Leader Election: Bully vs Ring

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1. Experimental Setup

10 nodes (IDs 1...10) each run as an independent *threading.Thread* with a private *queue.Queue* inbox. Nodes share **no** mutable state — the only inter-node interaction is enqueueing a *Message* on a peer's inbox, satisfying section 2.1's no-shared-memory rule. Every call to *Node.send()* sleeps for **50 ms** before delivery to simulate WAN latency (section 3, metric 3). Node 10 (the current leader) is killed before the run; a surviving node receives a *TRIGGER* message from the simulator and starts the election.

2. Measured Metrics (4 scenarios)

Algorithm	Scenario	Initiator	Messages	Hops	Wall-clock (ms)
Bully	Best-case	N9	10	2	2083.54
Bully	Worst-case	N1	90	3	2641.23
Ring	Best-case	N9	18	18	1060.01
Ring	Worst-case	N1	18	18	1071.22

Figure 1. Measured Messages, Hops and Wall-clock Time per scenario. Per-run message-by-message traces are in `logs/<algo>_<scenario>.log`; the raw numbers are also in `results/results.json`.

3. Closed-Form Complexity (sanity check)

Algorithm	Messages (best)	Messages (worst)	Hops	Time lower bound
Bully	$N = 10$	$O(N^2) \approx 90$	2 - 4	$2 * 50 \text{ ms} = 100 \text{ ms} + \text{OK-timeout}$
Ring	$2(N-1) = 18$	$2(N-1) = 18$	$2(N-1) = 18$	$2(N-1) * 50 \text{ ms} = 900 \text{ ms}$

Figure 2. Theoretical complexity for the failed-leader scenario ($N=10$, one dead node). Measured numbers match exactly: Bully best = 10 (1 ELECTION to N10 + 9 COORDINATOR broadcasts), Bully worst = 90 (45 ELECTION + 36 OK + 9 COORDINATOR; see `logs/bully_worst.log`), and Ring best = Ring worst = 18 (9 ELECTION around the ring + 9 COORDINATOR around the ring).

4. Analysis — Bully vs Ring

4.1 Message Complexity

Bully is **cheaper than Ring when the initiator is close to the top of the hierarchy** (10 vs 18 messages in the best case) because N9 only needs to probe N10, time out, and broadcast. As soon as the initiator is far from the top, Bully's recursive cascade explodes — every N_i receiving ELECTION starts its own election to all $N_j > i$, giving $\sum_{i=1}^{N-1} (N-i) = N(N-1)/2$ ELECTIONS, plus an OK for every ELECTION that hits an alive node, plus the leader's final COORDINATOR broadcast. Our measured 90 matches this $O(N^2)$ bound exactly. Ring is **scenario-independent**: it always sends $2(N-1) = 18$ messages regardless of who initiates, because the topology is fixed and each phase (ELECTION sweep, COORDINATOR sweep) traverses every alive node exactly once.

4.2 Network Hops (sequential depth)

Bully wins decisively here: just 2 hops best, 3 hops worst. The algorithm exploits parallelism — the initiator *broadcasts* ELECTION in one logical wave, the receivers reply OK *and* launch their own elections in a second wave, and the surviving highest-ID node broadcasts COORDINATOR in a third wave. No matter how big N gets, the depth stays a small constant. Ring's depth, by contrast, is **$2(N-1) = 18$ hops** for both scenarios, because the ring is fundamentally serial — each forward must wait for the previous one to arrive.

4.3 Wall-Clock Latency

With the 50 ms injected delay, hop count is the dominant term. Ring takes $\sim 1.06 \text{ s}$ (best) and $\sim 1.07 \text{ s}$ (worst), tracking the $18 \times 50 \text{ ms} = 900 \text{ ms}$ hop-floor closely. Bully's best-case wall-clock (2.08 s) is dominated by the 1.5 s OK-timeout (the price of waiting to confirm N10 really is dead) plus the 9-message COORDINATOR broadcast. Bully worst (2.64 s) is the same timeout plus a much larger COORDINATOR phase competing for thread time against the 90-message cascade. The take-away: **Bully's latency is paid in seconds of failure-detection timeout, Ring's is paid in serial hops** — on a high-RTT WAN the timeout matters less, on a fast LAN the hops matter less.

4.4 When to pick which

- **Bully** — small clusters, stable membership, topology that supports broadcast / full-mesh reachability, and where minimizing *turnaround rounds* matters more than minimizing message count (e.g. low-latency LANs in HPC / in-memory DB primaries).
- **Ring** — large or geographically distributed clusters where each node only knows its successor, where uniform / predictable message cost matters, and where you want the worst case to equal the best case (no "election storms"). The Chubby/ZAB family of real-world coordinator elections is closer in spirit to Ring + quorum than to Bully.
- **Failure-detection timeout** is the hidden cost in Bully: we used `OK_TIMEOUT_S = 1.5 s` because with the cascade a higher node's OK can be delayed by ~900 ms while it drains its inbox. Too short -> split-brain (multiple leaders); too long -> slow recovery. Ring has no such tuning knob.

5. Reproduction

```
python -m venv .venv && .venv\Scripts\activate.bat
pip install -r requirements.txt
python run_experiments.py # writes logs/ + results/
python report/generate_report.py # rebuilds this PDF
```